

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: **Data Analysis with SAS / SPSS / R**

PRACTICAL 3

AIM: Exploring data: View() or print() (R).

>COMMANDS

```
install.packages(c("readr", "psych"))
```

```
library(readr) # For efficient data reading  
library(psych) # For descriptive statistics  
# View the first few rows  
head(job_title)
```

```
# View the last few rows  
tail(job_title)
```

```
# Get the dimensions (rows and columns)  
dim(job_title)
```

```
# Get the structure (variable types and number of observations)  
str(job_title)
```

```
# See a summary of the dataset  
summary(job_title)
```

```
# Get the column names  
names(job_title)
```

```
# Use the 'psych' package for more detailed descriptive statistics  
describe(job_title)
```

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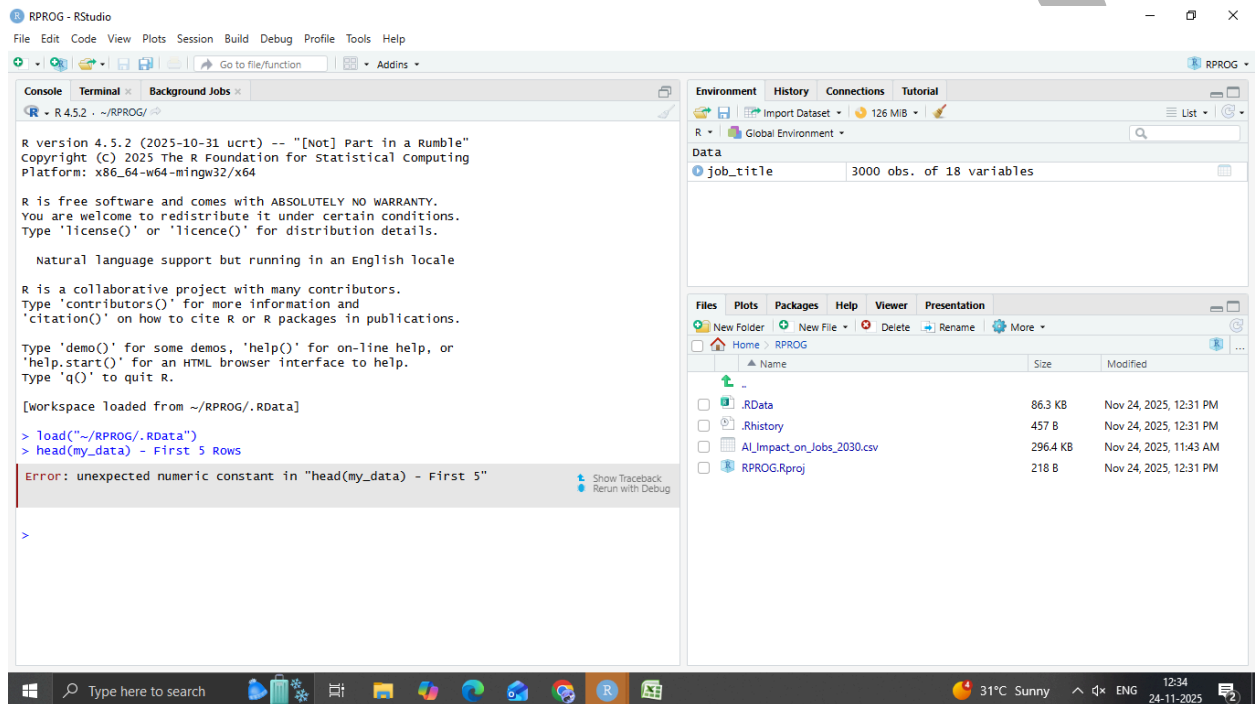
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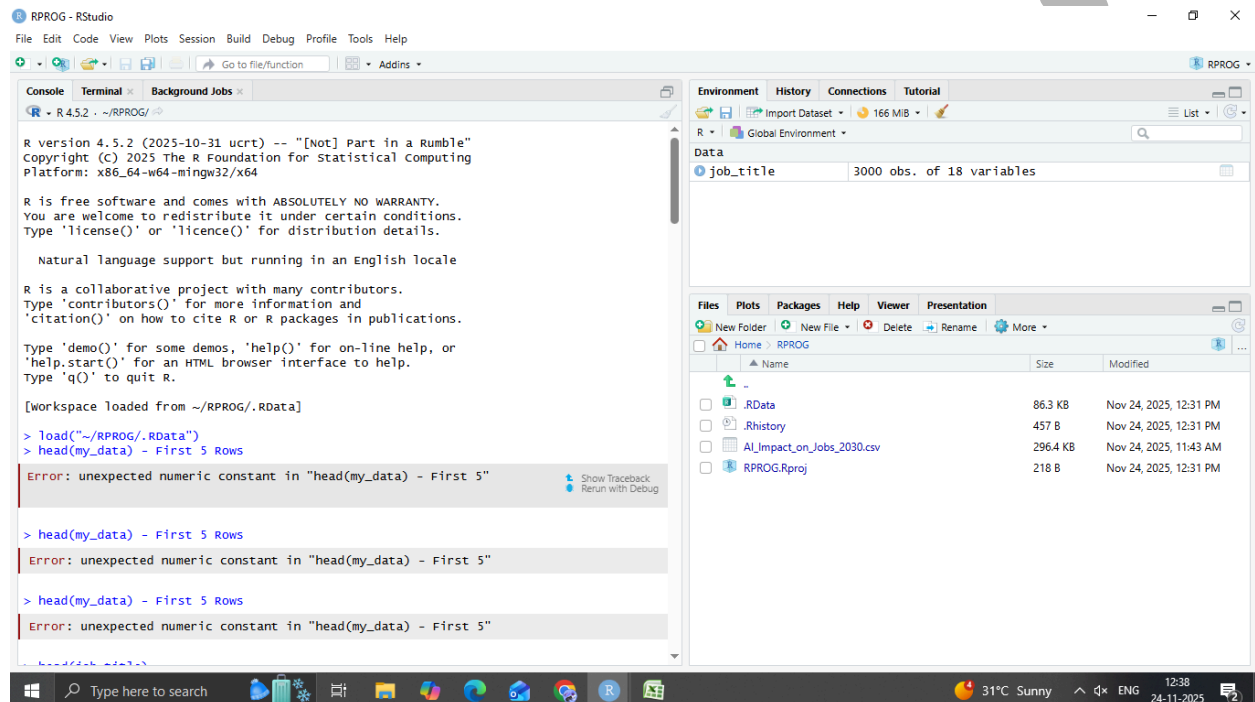


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The screenshot displays the RStudio environment. The console window on the left shows the R version 4.5.2 (2025-10-31 ucrt) and the workspace loaded from ~/RPROG/.RData. The file explorer on the right shows the project files: .RData (86.3 KB), .Rhistory (457 B), AI_Impact_on_Jobs_2030.csv (296.4 KB), and RPROG.Rproj (218 B). The console output includes the R license information and the command to load the workspace. The file explorer shows the project files and their sizes.

```
R version 4.5.2 (2025-10-31 ucrt) -- "[Not] Part in a Rumble"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[workspace loaded from ~/RPROG/.RData]
> load("~/RPROG/.RData")
> head(my_data) - First 5 Rows
Error: unexpected numeric constant in "head(my_data) - First 5"
> head(my_data) - First 5 Rows
Error: unexpected numeric constant in "head(my_data) - First 5"
> head(my_data) - First 5 Rows
Error: unexpected numeric constant in "head(my_data) - First 5"
```

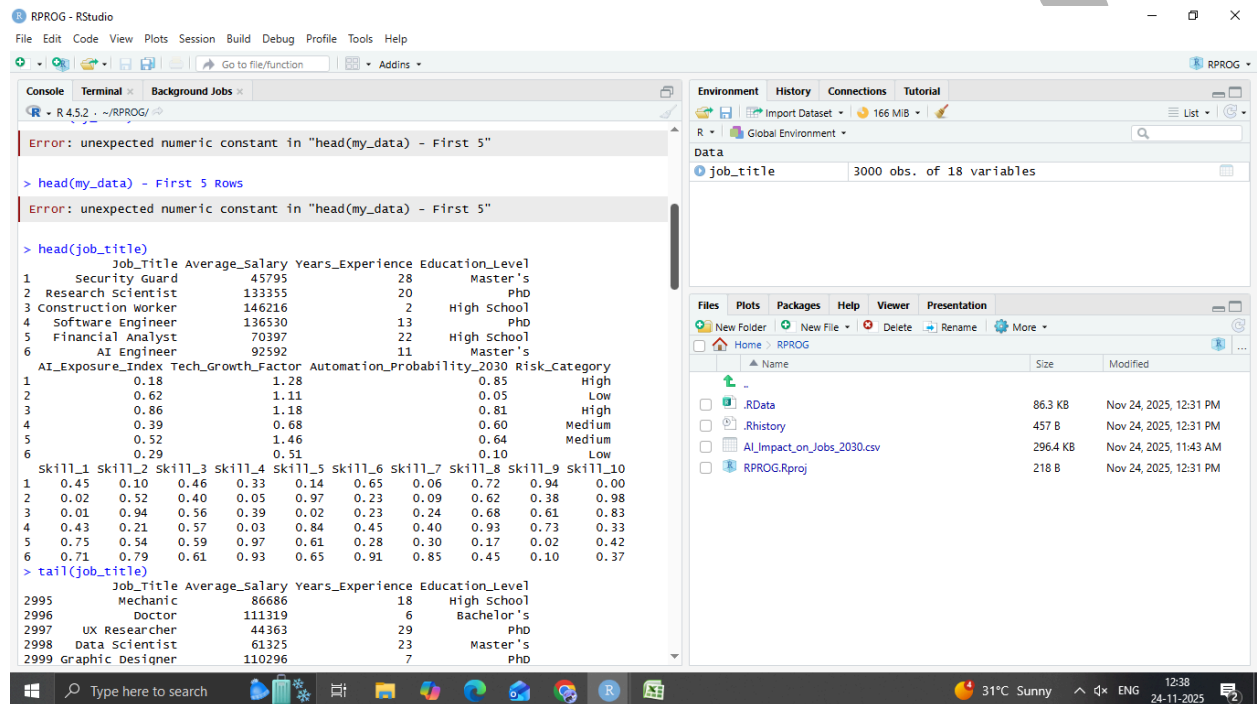
Name	Size	Modified
.RData	86.3 KB	Nov 24, 2025, 12:31 PM
.Rhistory	457 B	Nov 24, 2025, 12:31 PM
AI_Impact_on_Jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
RPROG.Rproj	218 B	Nov 24, 2025, 12:31 PM

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The screenshot shows the RStudio interface with the following content:

Console:

```
Error: unexpected numeric constant in "head(my_data) - First 5"

> head(my_data) - First 5 Rows

Error: unexpected numeric constant in "head(my_data) - First 5"

> head(job_title)
  Job_Title Average_salary Years_Experience Education_Level
1 Security Guard      45795          28      Master's
2 Research Scientist 133355          20      PhD
3 Construction Worker 146216           2 High School
4 Software Engineer  136530          13      PhD
5 Financial Analyst   70397          22 High School
6 AI Engineer        92592          11      Master's
AI_Exposure_Index Tech_Growth_Factor Automation_Probability_2030 Risk_Category
1      0.18           1.28           0.85      High
2      0.62           1.11           0.05      Low
3      0.86           1.18           0.81      High
4      0.39           0.68           0.60 Medium
5      0.52           1.46           0.64 Medium
6      0.29           0.51           0.10      Low
skill_1 skill_2 skill_3 skill_4 skill_5 skill_6 skill_7 skill_8 skill_9 skill_10
1  0.45  0.10  0.46  0.33  0.14  0.65  0.06  0.72  0.94  0.00
2  0.02  0.52  0.40  0.05  0.97  0.23  0.09  0.62  0.38  0.98
3  0.01  0.94  0.56  0.39  0.02  0.23  0.24  0.68  0.61  0.83
4  0.43  0.21  0.57  0.03  0.84  0.45  0.40  0.93  0.73  0.33
5  0.75  0.54  0.59  0.97  0.61  0.28  0.30  0.17  0.02  0.42
6  0.71  0.79  0.61  0.93  0.65  0.91  0.85  0.45  0.10  0.37

> tail(job_title)
  Job_Title Average_salary Years_Experience Education_Level
2995 Mechanic      86686          18 High School
2996 Doctor      111319           6 Bachelor's
2997 UX Researcher  44363          29      PhD
2998 Data Scientist  61325          23      Master's
2999 Graphic Designer 110296           7      PhD
```

Environment:

Global Environment

job_title 3000 obs. of 18 variables

Files:

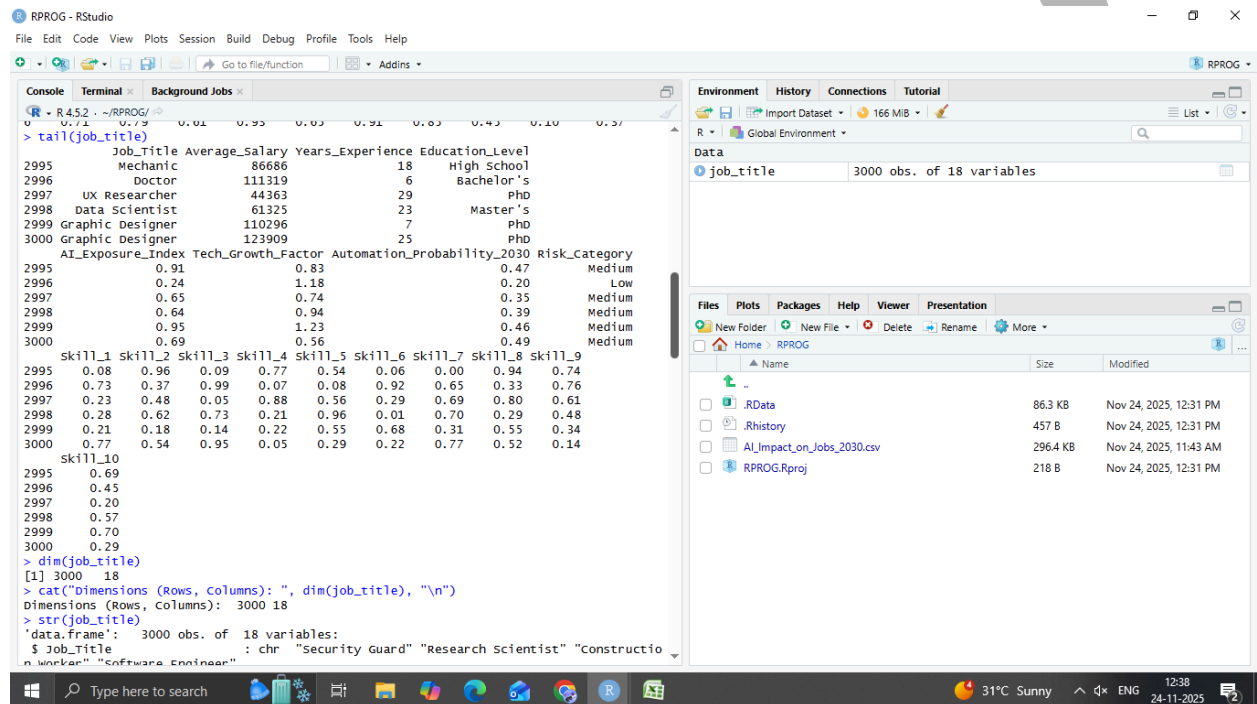
Name	Size	Modified
..		
.RData	86.3 KB	Nov 24, 2025, 12:31 PM
.Rhistory	457 B	Nov 24, 2025, 12:31 PM
AI_Impact_on_Jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
RPROG.Rproj	218 B	Nov 24, 2025, 12:31 PM

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The screenshot displays the RStudio environment with the following components:

- Console:** Shows the execution of `tail(job_title)` and `dim(job_title)`. The output for `tail(job_title)` displays a subset of the dataset with columns: Job_Title, Average_salary, Years_Experience, Education_Level, AI_Exposure_Index, Tech_Growth_Factor, Automation_Probability_2030, Risk_Category, and skill_1 through skill_9. The output for `dim(job_title)` shows dimensions of 3000 rows and 18 columns.
- Environment:** Shows the loaded data frame `job_title` with 3000 observations and 18 variables.
- Files:** Lists files in the current directory, including `.RData`, `.Rhistory`, `AI_Impact_on_Jobs_2030.csv`, and `RPROG.Rproj`.

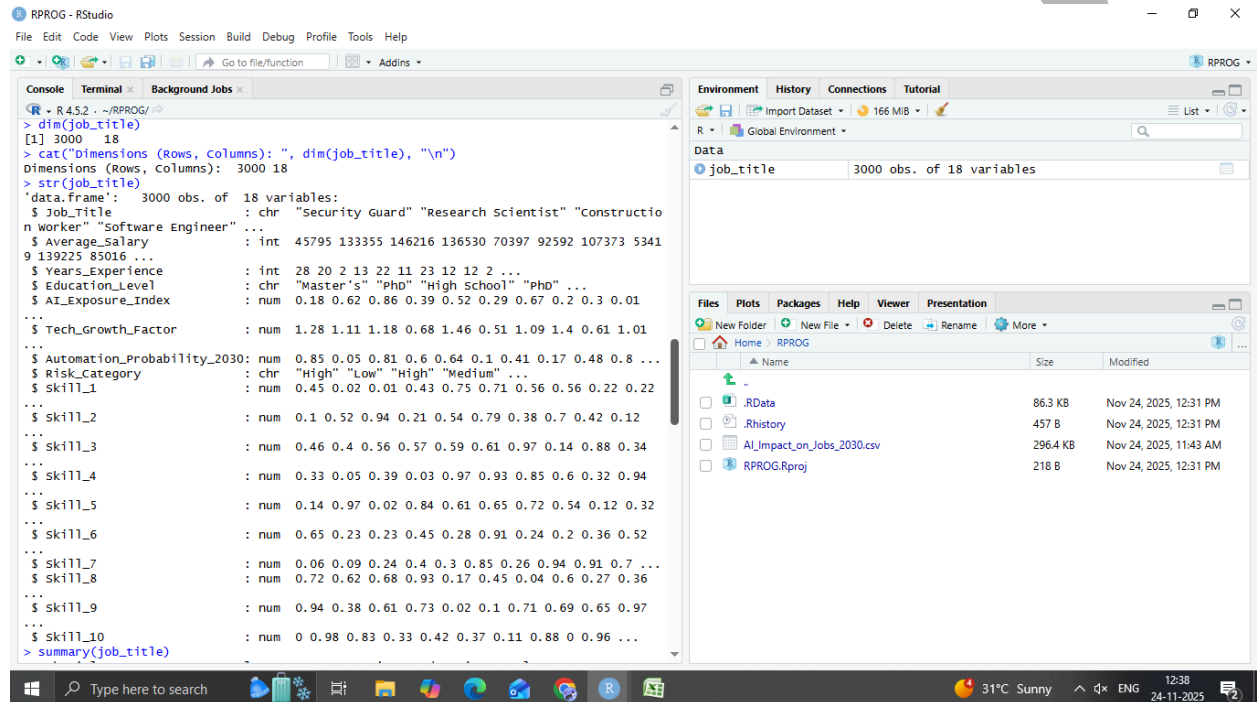
```
R - R 4.5.2 - ~/RPROG/
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Console Terminal Background Jobs
> tail(job_title)
  Job_Title Average_salary Years_Experience Education_Level
2995 Mechanic 86686 18 High School
2996 Doctor 111319 6 Bachelor's
2997 UX Researcher 44363 29 PhD
2998 Data Scientist 61325 23 Master's
2999 Graphic Designer 110296 7 PhD
3000 Graphic Designer 123909 25 PhD
  AI_Exposure_Index Tech_Growth_Factor Automation_Probability_2030 Risk_Category
2995 0.91 0.83 0.47 Medium
2996 0.24 1.18 0.20 Low
2997 0.65 0.74 0.35 Medium
2998 0.64 0.94 0.39 Medium
2999 0.95 1.23 0.46 Medium
3000 0.69 0.56 0.49 Medium
  skill_1 skill_2 skill_3 skill_4 skill_5 skill_6 skill_7 skill_8 skill_9
2995 0.08 0.96 0.09 0.77 0.54 0.06 0.00 0.94 0.74
2996 0.73 0.37 0.99 0.07 0.08 0.92 0.65 0.33 0.76
2997 0.23 0.48 0.05 0.88 0.56 0.29 0.69 0.80 0.61
2998 0.28 0.62 0.73 0.21 0.96 0.01 0.70 0.29 0.48
2999 0.21 0.18 0.14 0.22 0.55 0.68 0.31 0.55 0.34
3000 0.77 0.54 0.95 0.05 0.29 0.22 0.77 0.52 0.14
  skill_10
2995 0.69
2996 0.45
2997 0.20
2998 0.57
2999 0.70
3000 0.29
> dim(job_title)
[1] 3000 18
> cat("Dimensions (Rows, Columns): ", dim(job_title), "\n")
Dimensions (Rows, Columns): 3000 18
> str(job_title)
'data.frame': 3000 obs. of 18 variables:
 $ Job_Title : chr "Security Guard" "Research Scientist" "Constructio
n Worker" "Software Engineer"
```

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The screenshot displays the RStudio environment with the following components:

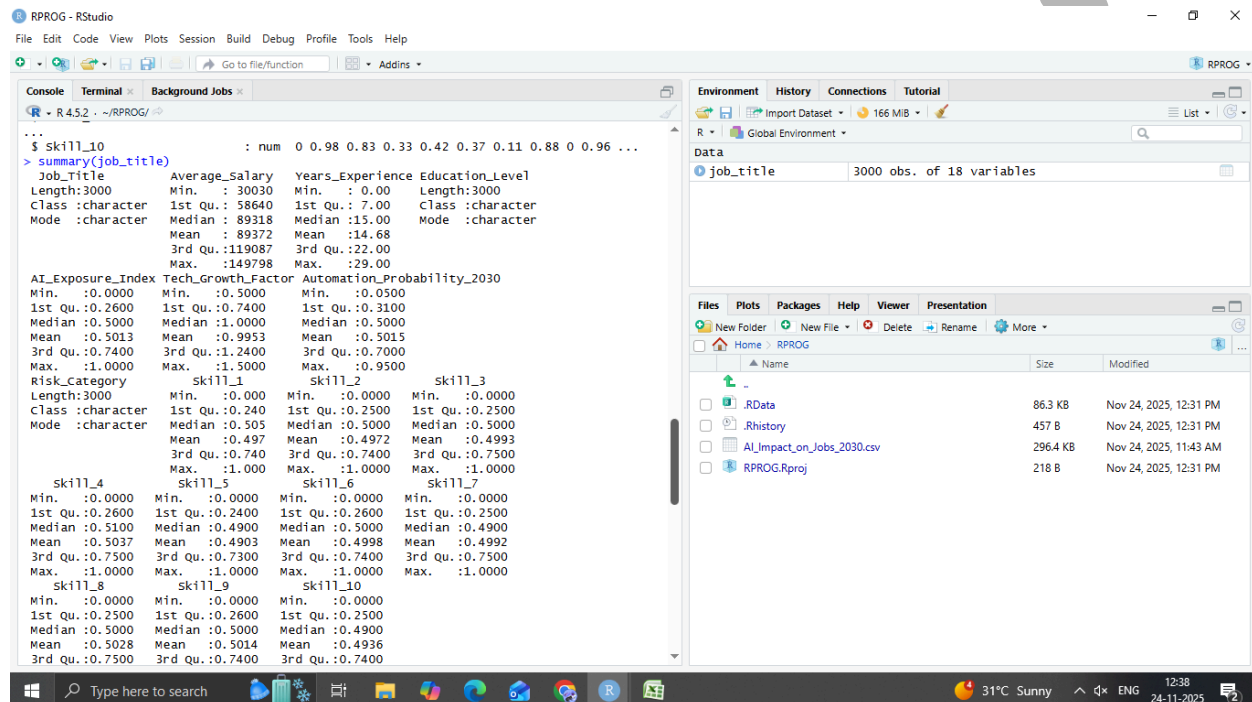
- Console:** Shows the execution of R code for loading and summarizing a dataset named 'job_title'. The code includes `dim(job_title)`, `cat("dimensions (Rows, columns): ", dim(job_title), "\n")`, `str(job_title)`, and `summary(job_title)`. The output indicates a data frame with 3000 observations and 18 variables.
- Environment:** Lists the loaded data objects: 'job_title' (3000 obs. of 18 variables).
- Files:** Shows the project files, including 'RData', 'Rhistory', 'AI_Impact_on_Jobs_2030.csv', and 'RPROG.Rproj'.
- Taskbar:** Displays the Windows taskbar with various application icons and system information (31°C Sunny, 12:38, 24-11-2025).

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```
$ skill_10 : num 0 0.98 0.83 0.33 0.42 0.37 0.11 0.88 0 0.96 ...
> summary(job_title)
Job_title      Average_Salary  Years_Experience  Education_Level
Length:3000    Min. : 30030      Min. : 0.00       Length:3000
Class :character 1st Qu.: 58640    1st Qu.: 7.00     Class :character
Mode :character  Median : 89318     Median :15.00     Mode :character
                Mean : 89372      Mean :14.68
                3rd Qu.:119087    3rd Qu.:22.00
                Max. :149798      Max. :29.00

AI_Exposure_Index Tech_Growth_Factor Automation_Probability_2030
Min. :0.0000      Min. :0.5000      Min. :0.0500
1st Qu.:0.2600    1st Qu.:0.7400    1st Qu.:0.3100
Median :0.5000    Median :1.0000     Median :0.5000
Mean :0.5013      Mean :0.9953      Mean :0.5015
3rd Qu.:0.7400    3rd Qu.:1.2400    3rd Qu.:0.7000
Max. :1.0000      Max. :1.5000      Max. :0.9500

Risk_Category    skill_1      skill_2      skill_3
Length:3000      Min. :0.000  Min. :0.0000  Min. :0.0000
Class :character 1st Qu.:0.240 1st Qu.:0.2500 1st Qu.:0.2500
Mode :character  Median :0.505  Median :0.5000  Median :0.5000
                Mean :0.497  Mean :0.4972   Mean :0.4993
                3rd Qu.:0.740 3rd Qu.:0.7400 3rd Qu.:0.7500
                Max. :1.000  Max. :1.0000   Max. :1.0000

skill_4      skill_5      skill_6      skill_7
Min. :0.0000  Min. :0.0000  Min. :0.0000  Min. :0.0000
1st Qu.:0.2600 1st Qu.:0.2400 1st Qu.:0.2600 1st Qu.:0.2500
Median :0.5100  Median :0.4900  Median :0.5000  Median :0.4900
Mean :0.5037   Mean :0.4903   Mean :0.4998   Mean :0.4992
3rd Qu.:0.7500 3rd Qu.:0.7300 3rd Qu.:0.7400 3rd Qu.:0.7500
Max. :1.0000   Max. :1.0000   Max. :1.0000   Max. :1.0000

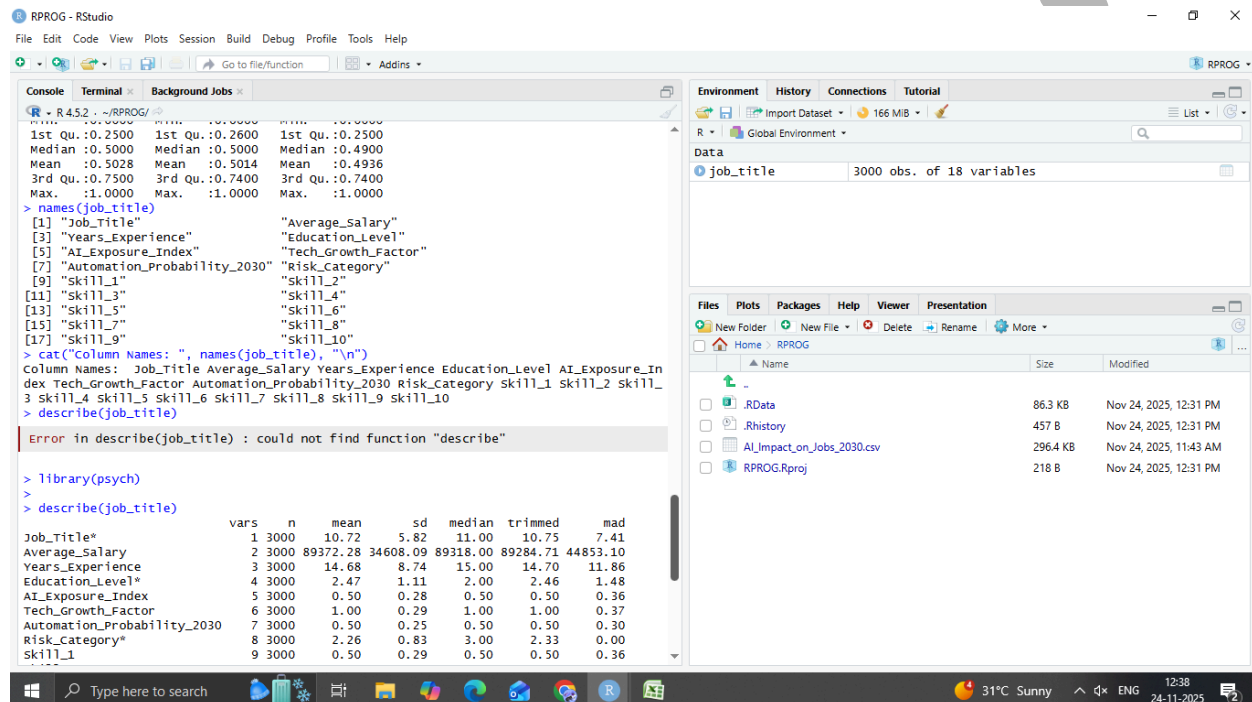
skill_8      skill_9      skill_10
Min. :0.0000  Min. :0.0000  Min. :0.0000
1st Qu.:0.2500 1st Qu.:0.2600 1st Qu.:0.2500
Median :0.5000  Median :0.5000  Median :0.4900
Mean :0.5028   Mean :0.5014   Mean :0.4936
3rd Qu.:0.7500 3rd Qu.:0.7400 3rd Qu.:0.7400
```

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The screenshot displays the RStudio environment. The console shows the following R commands and their outputs:

```
> names(job_title)
[1] "Job_Title" "Average_Salary"
[3] "Years_Experience" "Education_Level"
[5] "AI_Exposure_Index" "Tech_Growth_Factor"
[7] "Automation_Probability_2030" "Risk_Category"
[9] "skill_1" "skill_2"
[11] "skill_3" "skill_4"
[13] "skill_5" "skill_6"
[15] "skill_7" "skill_8"
[17] "skill_9" "skill_10"

> cat("Column Names: ", names(job_title), "\n")
Column Names: Job_Title Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor Automation_Probability_2030 Risk_Category skill_1 skill_2 skill_3 skill_4 skill_5 skill_6 skill_7 skill_8 skill_9 skill_10

> describe(job_title)
Error in describe(job_title) : could not find function "describe"

> library(psych)
> describe(job_title)
```

The Environment pane on the right shows the loaded data frame:

Variable	Count	Obs.	Vars.
job_title	3000	obs.	18 variables

The bottom pane shows the results of the `describe(job_title)` command using the `psych` package:

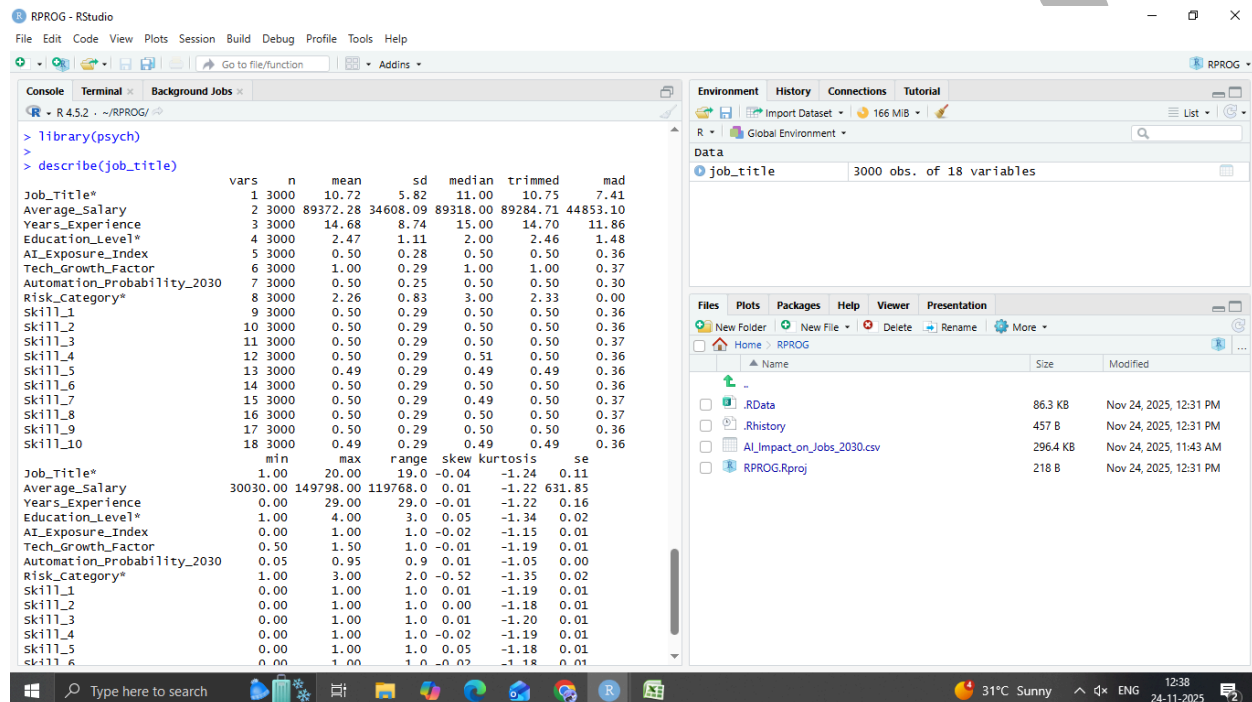
Variable	vars	n	mean	sd	median	trimmed	mad
Job_Title*	1	3000	10.72	5.82	11.00	10.75	7.41
Average_Salary	2	3000	89372.28	34608.09	89318.00	89284.71	44853.10
Years_Experience	3	3000	14.68	8.74	15.00	14.70	11.86
Education_Level*	4	3000	2.47	1.11	2.00	2.46	1.48
AI_Exposure_Index	5	3000	0.50	0.28	0.50	0.50	0.36
Tech_Growth_Factor	6	3000	1.00	0.29	1.00	1.00	0.37
Automation_Probability_2030	7	3000	0.50	0.25	0.50	0.50	0.30
Risk_Category*	8	3000	2.26	0.83	3.00	2.33	0.00
Skill_1	9	3000	0.50	0.29	0.50	0.50	0.36

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RStudio Console Output:

```
> library(psych)
> describe(job_title)
```

	vars	n	mean	sd	median	trimmed	mad
Job_Title*	1	3000	10.72	5.82	11.00	10.75	7.41
Average_Salary	2	3000	89372.28	34608.09	89318.00	89284.71	44853.10
Years_Experience	3	3000	14.68	8.74	15.00	14.70	11.86
Education_Level*	4	3000	2.47	1.11	2.00	2.46	1.48
AI_Exposure_Index	5	3000	0.50	0.28	0.50	0.50	0.36
Tech_Growth_Factor	6	3000	1.00	0.29	1.00	1.00	0.37
Automation_Probability_2030	7	3000	0.50	0.25	0.50	0.50	0.30
Risk_category*	8	3000	2.26	0.83	3.00	2.33	0.00
skill_1	9	3000	0.50	0.29	0.50	0.50	0.36
skill_2	10	3000	0.50	0.29	0.50	0.50	0.36
skill_3	11	3000	0.50	0.29	0.50	0.50	0.37
skill_4	12	3000	0.50	0.29	0.51	0.50	0.36
skill_5	13	3000	0.49	0.29	0.49	0.49	0.36
skill_6	14	3000	0.50	0.29	0.50	0.50	0.36
skill_7	15	3000	0.50	0.29	0.49	0.50	0.37
skill_8	16	3000	0.50	0.29	0.50	0.50	0.37
skill_9	17	3000	0.50	0.29	0.50	0.50	0.36
skill_10	18	3000	0.49	0.29	0.49	0.49	0.36

	min	max	range	skew	kurtosis	se
Job_Title*	1.00	20.00	19.0	-0.04	-1.24	0.11
Average_Salary	30030.00	149798.00	119768.0	0.01	-1.22	631.85
Years_Experience	0.00	29.00	29.0	-0.01	-1.22	0.16
Education_Level*	1.00	4.00	3.0	0.05	-1.34	0.02
AI_Exposure_Index	0.00	1.00	1.0	-0.02	-1.15	0.01
Tech_Growth_Factor	0.50	1.50	1.0	-0.01	-1.19	0.01
Automation_Probability_2030	0.05	0.95	0.9	0.01	-1.05	0.00
Risk_category*	1.00	3.00	2.0	-0.52	-1.35	0.02
skill_1	0.00	1.00	1.0	0.01	-1.19	0.01
skill_2	0.00	1.00	1.0	0.00	-1.18	0.01
skill_3	0.00	1.00	1.0	0.01	-1.20	0.01
skill_4	0.00	1.00	1.0	-0.02	-1.19	0.01
skill_5	0.00	1.00	1.0	0.05	-1.18	0.01
skill_6	0.00	1.00	1.0	-0.02	-1.18	0.01

Environment Pane:

- Global Environment
- job_title: 3000 obs. of 18 variables

Files Pane:

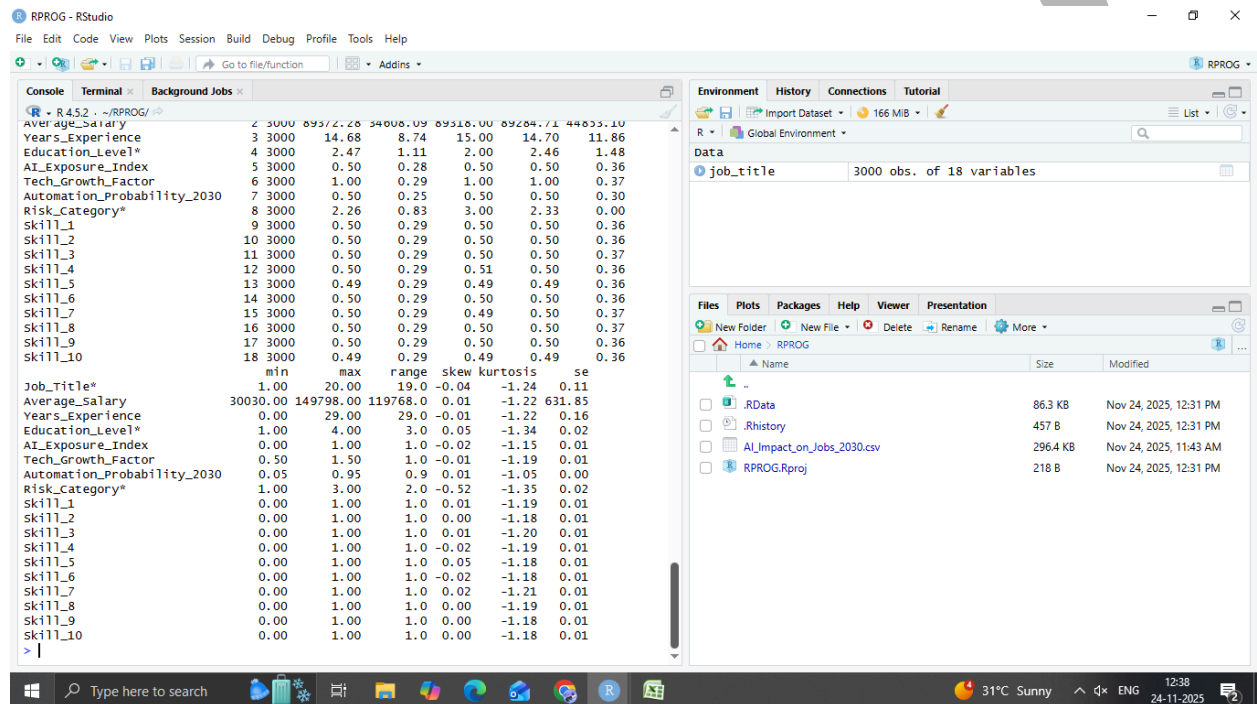
- Home > RPROG
- Files: .RData (86.3 KB, Nov 24, 2025, 12:31 PM), .Rhistory (457 B, Nov 24, 2025, 12:31 PM), AI_Impact_on_Jobs_2030.csv (296.4 KB, Nov 24, 2025, 11:43 AM), RPROG.Rproj (218 B, Nov 24, 2025, 12:31 PM)

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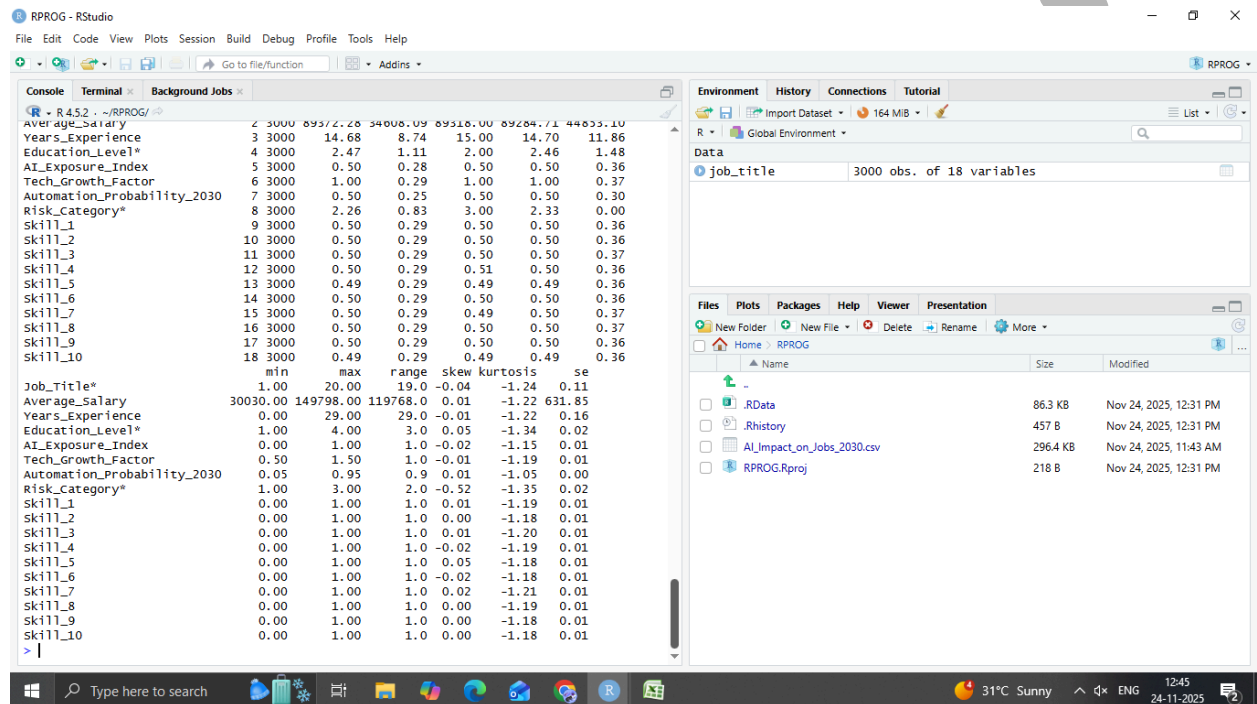


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